

BT503 Current Papers Spring 2020

1. Define Halophiles.	2
2. Define psychrophiles.	2
3. Define Biosensors.	2
4. Define co-metabolism.	2
5. How microbes are poly cyclic aromatic compounds.	2
6. Write two advantages of the conventional biological sludge dewatering process.	2
7. How plants are modified as bioindicator.	2
8. Mention two Natural sludge dewatering processes.	2
9. How quality of fruit should improve	2
10. Write two sources of dioxins?	2
11. What are xenobiotic?	2
12. Ex-situ technique?	2
13. What is Glyphosate	2
14. Define Bioscrubber	2
15. What is Sustainable bioremediation	2
16. Why DDT was banned?	2
17. Write two methods of Detergents bioremedies.	2
18. Consequences of OECD 2001 about global Biotechnology?	2
19. Define phytodegradation and name its forms?	2
20. What are living conditions of extremophiles? give examples	2
21. Difference between bioenhancement and biogumentation.	2
22. Difference between natural and conventional sludge dewatering process.	3
23. Difference between physical and mechanical sludge dewatering process.	3
24. Name enzyme, solution and equipment's used in the genetic engineering.	3
25. There classes of biosensor and two advantages and disadvantages of immunsensor.	3
26. Enlist the types of solid waste.	3
27. Phytoplankton?	3
28. How quality of crops should enhance?	3
29. Write 3 oxidation products of TNT.	3
30. Process of thickening of sludge.	3
31. Write features of phenols that make it pollutant?	3
32. What is T-DNA?	3
33. What is Environmental biotechnology	3
34. methods to get a good quality crop	3
35. biological remediation.	3
36. Advantages of association of fungi with plants	3
37. With help of named example discuss how whole organisms can be used the measure potential biological Toxicity of water soil sample?	3
38. Describe importance of anaerobic zone in biological process of phosphorus removal in activated sludge treatment?3	
39. Describe natural process to remove pathogenic environment.	5
40. Environmental concern to the release of the hormone into environment.	5
41. Role of filter medium in biofilter also name the compound of the filter medium.	5
42. Five impact of chlorobenzene & dioxin on human health.	5
43. write a note on cDNA libraries	5
44. Write the five properties of soil.	5
45. Write five advantages of Biofilter	5
46. Explain Vitrification methods	5

47. Why DDT banned reason of its banned although it is useful method for removal of insects? 5
48. Genetic modification a popular plant is used for removal of mercury. describe how? 5
49. Factors affecting the techniques of Insitu and exsitu. 5
50. What are applications of biosensors in detection of biocides? 5
51. Explain role of enzyme, solution & equipment in genetic engineering? 5
52. Explain genomic libraries? 5
53. solid waste steps. 5
54. What are environment factor that effect bioremediation briefly describes? 5
55. List any five source of solid waste? 5
56. What are PCBS? Which biosensors detect them? 5
57. Mention some metallic coagulant for sludge dewatering? 5
58. Phytovolatilisation 10
59. Dental not sludge treatment stage before final disposal steps. 10
60. Describe biosensors single transduction and bio recognition principle. 10
61. Write the procedure of Sludge treatment 10
62. Explain western and Northern blotting. 10
63. Cloning Vector. 10
64. Briefly describe different categories of Solid waste 10
65. What are risks of Solid Wastes regarding transfer of disease? 10

MCQS:

1. What is not true about bio scrubbers?.....**Remove toxic metals.**
2. Following is true about imhoff tank used in anaerobic filter system?.....
.....**Can be a single-chamber tank or a two-compartment tank.**
3. Explosives of characterized by presence of _ group?.....**Nitro.**
4. The term "endophytes" is usually used to denote microorganisms?.....**Non-Pathogenic.**
5. UASB stands for?.....**Upflow anaerobic sludge blanket (UASB) reactors.**
6. Facultative ponds which receive raw sewage is also known as? **Primary ponds.**
7. The bio scrubber is not itself truly a biological treatment system, but rather a?
.....**Highly efficient method of removing odour components by dissolving them.**
8. Biodiesel is not derived from?**Algae.**
9. An aquatic plant?**Macrophyte.**
10. Important factors to be considered when evaluating and selecting unit operations and processes except?
.....**Sterilization of vessels.**
11. Dioxins released as by-products in a number of chemical processes?.....**Do not pose threats to environment.**
12. Nonspecific enzymes are?**oxygenase's.**
13. Wastewater is applied to the soil, supplying water and nutrients necessary for plant growth. Which is not true?
.....**Generation of aerosols.**
14. Which is not used regarding desulphurization of coal?**Psychrophiles.**
15. Suitable temperature is?**20—30C.**
16. Statement about phytoremediation is false?
17. What is false about Collection stage?**This is the stage when solid waste is transported to the final disposal site.**

18. False statement about on site disposal of waste? **Waste volumes are large.**
19. Which of following does not use to make vessel of biofilter? **Formic Sheet.**
20. Preliminary treatment involves the removal of?..... **Coarse Solid.**
21. In bacteria genetic transfer of materials occurs through? **Plasmid.**
22. Wolffia globosa is an important tool for?..... **Cadmium.**
23. Optimum PH for local environment conditions lies in the range?..... **6.5—7.5.**
24. _____ uses High temperature for fuse contaminated material..... **Vitrification**
25. A step in the development of biosensors is?.....
..... **The immobilization of the biological component at the transducer surface.**
26. Associations of fungi with roots of vascular plants? **Mycorrhizae.**
27. The ability of a single compound to be degraded can be affected by the presence of other contaminants?
..... **Cometabolism.**
28. Which is true for Environment Biotechnology? **enhance or optimize conditions for existing biological systems to make their activities happen faster or more efficientl'ß**
29. Thermophile thrive at temperatures above? **45C.**
30. _____ are compounds which are man-made..... **Xenobiotic**
31. Natural estrogens are deactivated in humans by? **Glucuronidation.**
32. Collection of total Genomic DNA from an organism? **Genomic libraries.**
33. _____ refers to how pollutants enter a food chain..... **Bioaccumulation**
34. Which statement about Bioscrubber is FALSE?..... **They are useful for the removal of toxic metals**
35. Which of the following is NOT a possible destination of the pollutant in the soil?..... **appearance in surrounding air**
36. The presence of metals in environment may be detected visually by the use of bacteria. One such example is *Vogesella indigofera* that gives an idea of metal contamination by producing..... **Coloured pigments**
37. Which statement about chlorinated aliphatic hydrocarbons is correct?..... **They usually act as proton donor**
38. In the conventional activated sludge system. continuous arrival of food in the form of BOD to the aeration tank, leads to rapid and continuous bacterial growth. All of the following problems caused by increased bacterial growth. Except..... **the overall efficiency of the process is enhanced**
39. Which kind of the technique are suitable for bioremediation?..... **In-situ**
40. Unlike other marine species, _____ are characteristically untroubled by biofouling and study of their skin have established that its unique texture coupled with its inherent antimicrobial properties help keep them clean of fouling organisms..... **Sharks**
41. Which plant is used as a model to protect from various agents?..... **Tobacco plant**
42. The airflow in the submerged aerated biofilter is always _____ while the liquid flow can be _____
..... **upflow, downflow**
43. Which of the following approaches do not remove the pollutants but only immobilize the pollutants?
..... **phytostabilization**

44. The ____ product of pollutant may be more harmful than the original pollutant itself, therefore it is important to understand the chemistry of the pollutant.....**breakdown**
45. Among the first synthetic organic chemicals to create environmental problems were ____.....**Detergents**
46. Immobilization refer to the removal of contaminants, typically metals, by means of ____ by various microbes or plants species.....**adsorption**
47. Which of the following factors has no effect on the execution and implementation of environmental technology?
.....**health of the general population**
48. Dioxins have many congeners. The most toxic is ____ which is thought to be the most poisonous synthetic chemical.....**TCDD (tetrachlorodibenzo-p-dioxin)**
49. How much arsenic can be accumulated by using Pteris vittata?.....**2-3%**
50. Which of the following methods are best suited to instances of relatively localized pollution within a site, typically in 'hot-spots' of medium to relatively high concentration which are fairly near to the surface.....**Ex-Situ**
51. For land based applications, the soil type best suited biotechnological inventions are ____.....**Sand and Gravel**
52. Plants containing the novel DNA grow, a gene coding is frequently for ____..... **all of the above**
53. Deliberate transfer of a gene from one organism to another is called ____.....**Foreign gene**
54. Western blotting technique is used for the detection of ____.....**Protein**
55. The term "endophytes" is usually used to denote ____ microorganism.....**non-pathogenic**
56. Which of the following have the innate ability to degrade hydrocarbons contaminants through metabolic action and biosurfactant production..... **hydrocarbonoclastic bacteria (HCB)**
57. What is false about poor biodegradation of hydrocarbons?..... **Easily degraded by enzymatic attack**
58. Liquid biofuels are of considerable importance due to
59. Which process used in the natural process of sludge dewatering
60. In aerobic Cometabolism of CAHs ---- is required ----